

Learning how to prove Workshop Week 4

Permutations



Cycles

1. The **order of a permutation** $\sigma \in S_n$ is the smallest positive integer m such that $\sigma^m = \varepsilon$.
 - (a) Suppose that $\sigma \in S_n$ has order m . If $\ell \geq 1$ is such that $\sigma^\ell = \varepsilon$, then $m|\ell$.
 - (b) If σ is an r -cycle, prove that then σ has order r . In other words, the order of a cycle equals its length.
 - (c) If $\sigma = \gamma_1 \dots \gamma_s$ is a product of disjoint r_i -cycles γ_i then σ has order $m = \text{lcm}(r_1, \dots, r_s)$.
 - (d) Use (c) to find the order of the following permutations:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 5 & 7 & 9 & 14 & 10 & 11 & 12 & 8 & 3 & 13 & 2 & 6 & 4 & 1 \end{pmatrix}$$

$$\tau = (1 \ 2)(3 \ 4 \ 5 \ 6)$$

Solution: We are using a common technique/trick/method to solve (a). When we are asked to prove that $a|b$ (for a, b integers), where a has been defined as the least positive integer satisfying a property (P). Then the idea is to apply the Division Algorithm to write b as $b = qa + r$; since $0 \leq r < a$, if we prove that r has the property (P), then we will obtain that $r = 0$, since a (by definition) was the smallest positive integer with that property.

(a) Let $\sigma \in S_n$ of order m and $\ell \geq 1$ such that $\sigma^\ell = \varepsilon$. Applying the Division Algorithm we have that

$$\ell = qm + u, \quad \text{for } q \in \mathbb{Z} \text{ and } u \in \mathbb{Z} \text{ with } 0 \leq u < m.$$

If we prove that $u = 0$, then $\ell = qm$; which says that $m|\ell$. To do so, we will show that $\sigma^u = \varepsilon$. From $\ell = qm + u$ we get that $u = \ell - qm$, which yields that

$$\sigma^u = \sigma^{\ell - qm} = \sigma^\ell \cdot (\sigma^m)^{-q} = \varepsilon,$$

since $\sigma^\ell = \varepsilon$ and $\sigma^m = \varepsilon$. Since u satisfies that $\sigma^u = \varepsilon$, and m is the smallest positive integer with that property; we get that $u = 0$, since $u < m$.

(b) Suppose that $\sigma = (k_1 \ k_2 \ \dots \ k_{r-1} \ k_r) \in S_n$. Then:

$$\begin{aligned} \sigma(k_1) &= k_2, \\ \sigma^2(k_1) &= \sigma(k_2) = k_3, \\ \sigma^3(k_1) &= \sigma(k_3) = k_4, \\ &\vdots \\ \sigma^{r-1}(k_1) &= \sigma(k_{r-1}) = k_r, \\ \sigma^r(k_1) &= \sigma(k_r) = k_1. \end{aligned}$$

From here, we obtain that r is the smallest positive integer such that $\sigma^r(k_1) = k_1$. Rewriting σ as $\sigma = (k_2 \ k_3 \dots k_{r-1} \ k_r \ k_1)$ and reasoning as before, we get that r is the smallest positive integer such that $\sigma^r(k_2) = k_2$. Following this process, we end up proving that r is the smallest positive integer such that $\sigma^r = \varepsilon$. Therefore, the order of σ is r , which is the length of σ .

(c) Suppose that $\sigma = \gamma_1 \gamma_2 \dots \gamma_s$, where each γ_i is an r_i -cycle. Let t be the order of σ and $m = \text{lcm}(r_1, r_2, \dots, r_s)$. Take $j \in \{1, \dots, s\}$ and $x \in \gamma_j$. First of all notice that x is fixed for all γ_i with $i \neq j$ since the γ_i 's are disjoint cycles, by hypothesis. From here we get that

$$\sigma^k(x) = (\gamma_1 \dots \gamma_s)^k(x) = \gamma_j^k(x), \quad \text{for all } k \geq 1. \quad (1)$$

On the other hand, since t is the order of σ we have that

$$x = \sigma^t(x) = (\gamma_1 \dots \gamma_s)^t(x) = \gamma_j^t(x),$$

for every element $x \in \gamma_j$. Thus $\gamma_j^t = \varepsilon$, which implies that $r_j | t$, since r_j is the order of γ_j by (a). Thus $m | t$ since $m = \text{lcm}(r_1, r_2, \dots, r_s)$.

On the other hand, since $r_j | m$ we have that $\gamma_j^m(a) = a$, for all $a \in X_n$, and from (1) we obtain that $\gamma_j^m(a) = a$, for all $a \in X_n$. It implies that $\sigma^m = \varepsilon$, which yields that $t | m$ since t is the order of σ .

We have shown that $t | m$ and $m | t$, and both t and m are positive integers, so necessarily $t = m$.

(d) Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 5 & 7 & 9 & 14 & 10 & 11 & 12 & 8 & 3 & 13 & 2 & 6 & 4 & 1 \end{pmatrix}$. In order to apply (b), we first factorize σ into disjoint cycles:

$$\sigma = (1 \ 5 \ 10 \ 13 \ 4 \ 14)(2 \ 7 \ 12 \ 6 \ 11)(3 \ 9)$$

Therefore, the order of σ is $\text{lcm}(6, 5, 2) = 30$.

The permutation τ is already a product of disjoint cycles, so an application of (b) gives that the order of τ is $\text{lcm}(2, 4) = 4$.